Practice Exam - 1

Question 13 of 117

A 72-year old man presents for evaluation of persistent fatigue and lower extremity edema. The patient was recently hospitalized for heart failure requiring intravenous diuretics. He continues to notice swelling in his legs and feels dizzy when standing up.

His past medical history is notable for lumbar spinal stenosis, carpal tunnel syndrome, and aortic stenosis status-post transcatheter aortic valve replacement a few months earlier. In the office, blood pressure seated is 100/60 mm Hg with heart rate of 65 bpm. With standing, blood pressure drops to 80/60 mm Hg with heart rate 90 bpm.

Transthoracic echocardiography shows preserved left ventricular ejection fraction with interventricular septum of 15 mm and posterior wall thickness of 14 mm. The aortic valve is well-seated and gradients across the valve are normal.

Laboratory testing reveals a sodium 128 mmol/L, potassium 3.7 mmol/L, creatinine 2.2 mg/dL. Albumin 3.2 g/dL. Kappa free light chains are 1.8 mg/dL with lambda free light chains of 1.2 mg/dL.

Which of the following is the next best diagnostic test?

A. Cardiac magnetic resonance imaging

B. Endomyocardial biopsy

C. Technetium pyrophosphate scintigraphy

D. Cardiac position emission tomography

E. Genetic Testing

Commentary References

Testing Point: The Board-certified AHFTC specialist should recognize the presentation of ATTR amyloidosis and non-invasive methods for diagnosis.

Correct Answer: C. Technetium pyrophosphate scintigraphy

Rationale: Technetium pyrophosphate scintigraphy, also known as a PYP scan, is a non-invasive test used in the diagnosis of ATTR amyloidosis. The patient's clinical presentation (unexplained increased LV wall thickness together with persistent heart failure symptoms and orthostatic hypotension) including associated conditions such as carpal tunnel syndrome, lumbar spinal stenosis, and aortic stenosis are all associated with ATTR amyloidosis. Given that the patient has a normal serum k/λ free light chain ratio, AL amyloidosis is much less likely. The use of PYP scans has dramatically reduced the need for endomyocardial biopsies which are more invasive as a first diagnostic test (Answer B).

Answer D: Cardiac positron emission tomography is more helpful in evaluation for cardiac sarcoidosis or myocarditis, which is not consistent with this patient's history. Answer A: While cardiac magnetic resonance imaging will provide additional information, the PYP scan is a more specific test and a better choice given the high pretest probability for ATTR.

Answer E: This is not the best next step because it may fail to detect non-hereditary transthyretin amyloidosis (wild-type ATTR).